DATA ANALYTICS PORTFOLIO CASE STUDY

E-Commerce Sales & Customer Analytics

An end-to-end analysis using Microsoft Excel, MySQL and Power BI

TOOLS	DATASET	SCOPE
Excel • MySQL • Power BI	Sample Superstore	9,994 rows • 21 columns

A practical business analytics case study demonstrating data preparation, SQL analysis, KPI development, interactive dashboarding and evidence-based recommendations.

Executive Summary

This project analyses the Sample Superstore retail dataset to understand sales performance, profitability, customer segments, regional contribution, product performance and shipping patterns. Excel supports inspection and exploratory analysis, MySQL provides structured analytical queries, and Power BI presents the findings across three dashboard pages.

KPI	RESULT	DEFINITION
Total sales	$2,297,200.86	Sum of Sales
Total profit	$286,397.02	Sum of Profit
Orders	5,009	Distinct Order IDs
Unique customers	793	Distinct Customer IDs
Units sold	37,873	Sum of Quantity
Profit margin	12.47%	Profit ÷ Sales
Average order value	$458.61	Sales ÷ distinct orders

The main worksheet contains 9,994 transaction-line records and 21 columns, covering 03 Jan 2014 to 30 Dec 2017. Direct checks found 0 blank cells and 0 exact duplicate rows.

Headline findings

- Technology leads sales ($836,154.03) and profit ($145,454.95).
- West leads regional sales ($725,457.82) and profit ($108,418.45).
- Consumer is the largest customer segment by sales ($1,161,401.34).
- Standard Class is the leading shipping mode by sales ($1,358,215.74).
- Tables is the lowest-profit sub-category ($-17,725.48); Bookcases and Supplies are also loss-making.

Business problem and objectives

Retail decision-makers need a consistent view of revenue drivers, profit contribution, customer value and potential margin leakage. This analysis asks which categories and regions lead, how segments contribute, which products are profitable or loss-making, and how sales vary over time and by shipping mode.

- Compare sales and profit by category, sub-category and region.
- Measure segment contribution using sales, profit and distinct orders.
- Rank customers, products and states by relevant metrics.
- Review monthly sales patterns and shipping-mode mix.
- Identify discount and product-profitability areas for further investigation.

Dataset overview

The main worksheet is “Sample - Superstore”. Each row represents a product line within an order, not a unique order.

FIELD GROUP	FIELDS	USE
Order	Row ID, Order ID, Order Date, Ship Date, Ship Mode	Order counts, trends and fulfilment
Customer	Customer ID, Customer Name, Segment	Customer and segment analysis
Geography	Country, City, State, Postal Code, Region	Regional comparisons
Product	Product ID, Category, Sub-Category, Product Name	Product mix and rankings
Measures	Sales, Quantity, Discount, Profit	Revenue, volume and profitability

Date coverage: 03 Jan 2014 – 30 Dec 2017. The source is a U.S. retail sample.

Tools and workflow

TOOL	APPLICATION IN THIS PROJECT
Excel	Data inspection, pivot tables, calculations, charts and summary insights.
MySQL	SELECT queries, aggregation, GROUP BY, ORDER BY, distinct counts, date parsing and rankings.
Power BI	DAX measures, KPI cards, category/customer/product visuals and interactive slicers.

Data preparation and quality checks

- Inspected worksheet structure, fields and sample records.
- Confirmed 9,994 rows and 21 columns in the main sheet.
- Workbook dates are typed as dates and support time-based analysis.
- Main-sheet check: 0 null cells and 0 exact duplicate rows.
- Core measures use SUM for Sales, Profit and Quantity; distinct counts for Order ID and Customer ID.

Scope note: these quality checks apply to the uploaded Excel source sheet. They do not establish that the MySQL import or Power BI model contains exactly the same rows. The submitted SQL file does not include an executed row-count reconciliation.

Excel analysis

The workbook includes dedicated sheets for category and regional sales, category and regional profit, customer segments, monthly sales, shipping method, profitable products, states, sub-categories, quantity, average discount, profit margin, summary and project insights.

ANALYSIS	FINDING
Category	Technology is the largest category by sales and profit.
Region	West ranks first in sales and profit.
Segment	Consumer leads sales and profit contribution.
Shipping	Standard Class is the largest shipping mode by sales.
Monthly trend	November 2017 is the strongest individual month.

SQL analysis

The uploaded SQL script contains 29 SELECT statements by keyword count. The statements cover KPI totals, category and region summaries, monthly and annual trends, shipping, segments, customer rankings, product rankings and profitability.

PATTERN	PURPOSE	BUSINESS QUESTION
SUM() + GROUP BY	Aggregate by category, region or segment	Where is value generated?
COUNT(DISTINCT ...)	Count unique orders	How much activity by segment?
STR_TO_DATE(), YEAR(), MONTH()	Parse and group order dates	How does performance change over time?
ORDER BY + LIMIT	Rank top/bottom products, states and customers	Who leads or needs attention?
AVG(Discount)	Compare average discounts	Where is discounting highest?

Representative query:

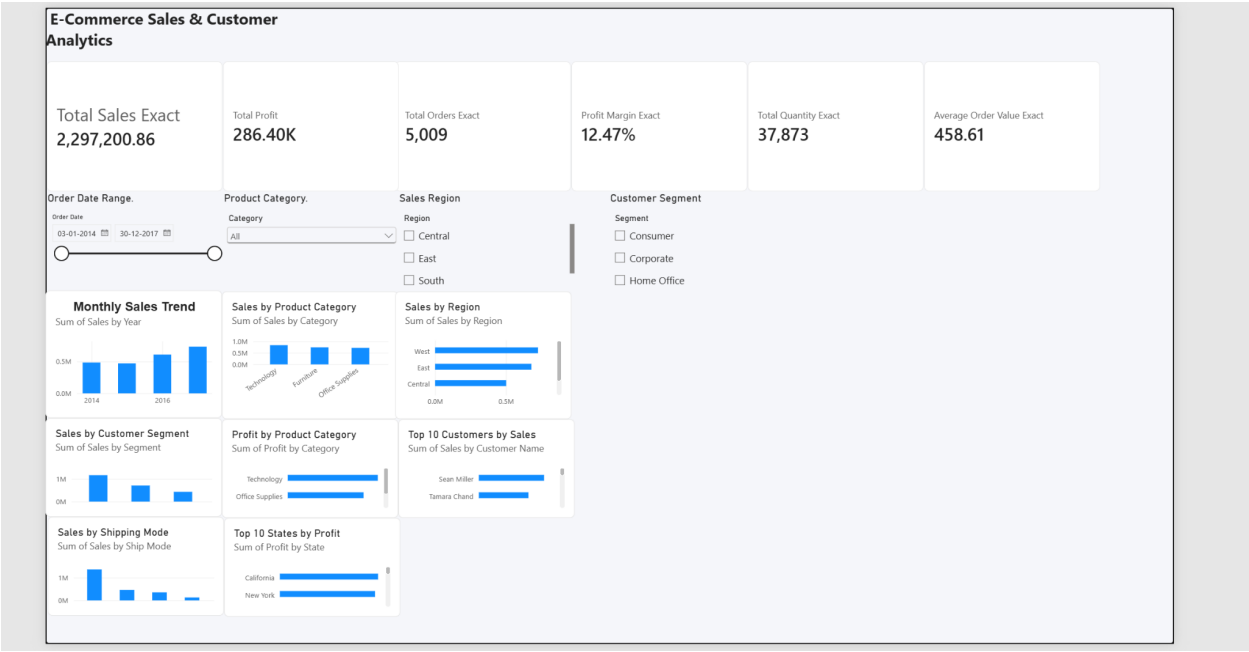
```
SELECT Category, SUM(Sales) AS Total_Sales
FROM `ecommerce_analytics`.`sample - superstore`
GROUP BY Category
ORDER BY Total_Sales DESC;
```

This query aggregates sales by category and sorts the result from highest to lowest. The SQL file demonstrates query design, but no executed output was supplied; validate SQL results after confirming the database import is complete.

Power BI dashboard analysis

The three supplied screenshots are included below. The PBIX file was supplied, but its internal model was not independently extracted; the descriptions therefore reflect the visible screenshots and cross-checkable workbook figures.

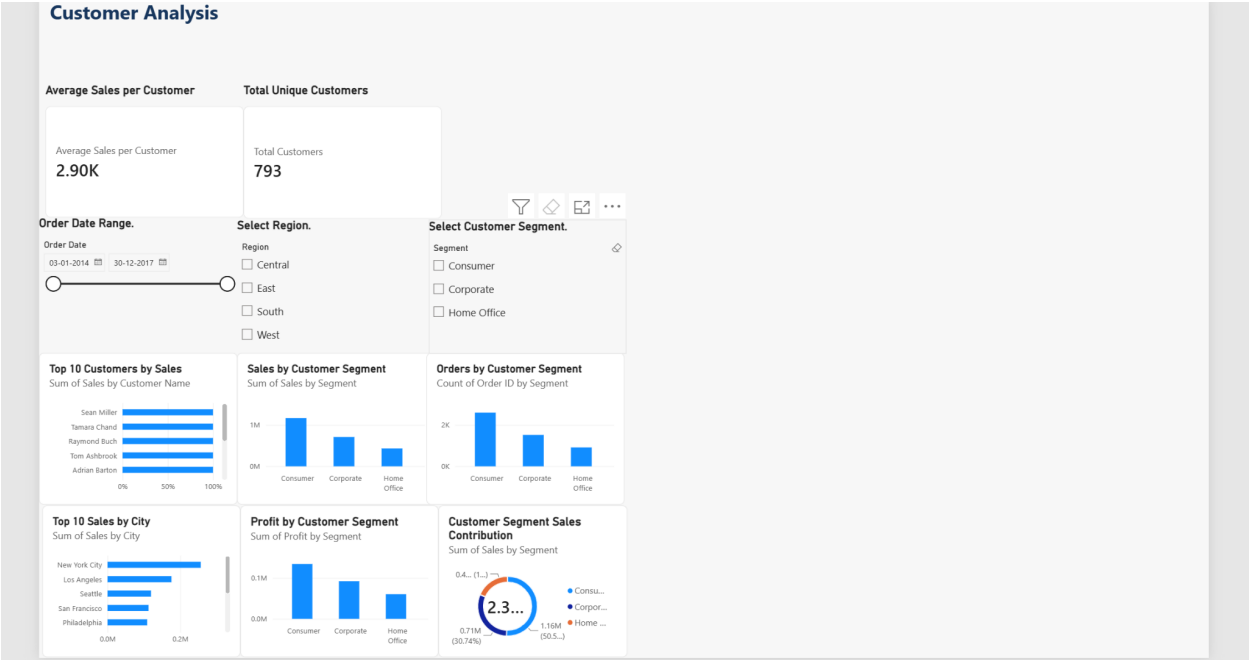
Page 1 — Sales Overview



The page displays six KPIs: sales $2,297,200.86, profit $286,397.02, 5,009 orders, 12.47% profit margin, 37,873 units and $458.61 average order value. Date, category, region and segment slicers filter the monthly trend, category/region/segment charts, category profit, top customers, shipping mode and state profit visuals.

Layout improvement: the screenshot shows visuals concentrated in the left portion of a wide canvas, leaving substantial unused space on the right. Reflow or resize the visuals, reduce the canvas width, and consider removing internal chart scrollbars where practical.

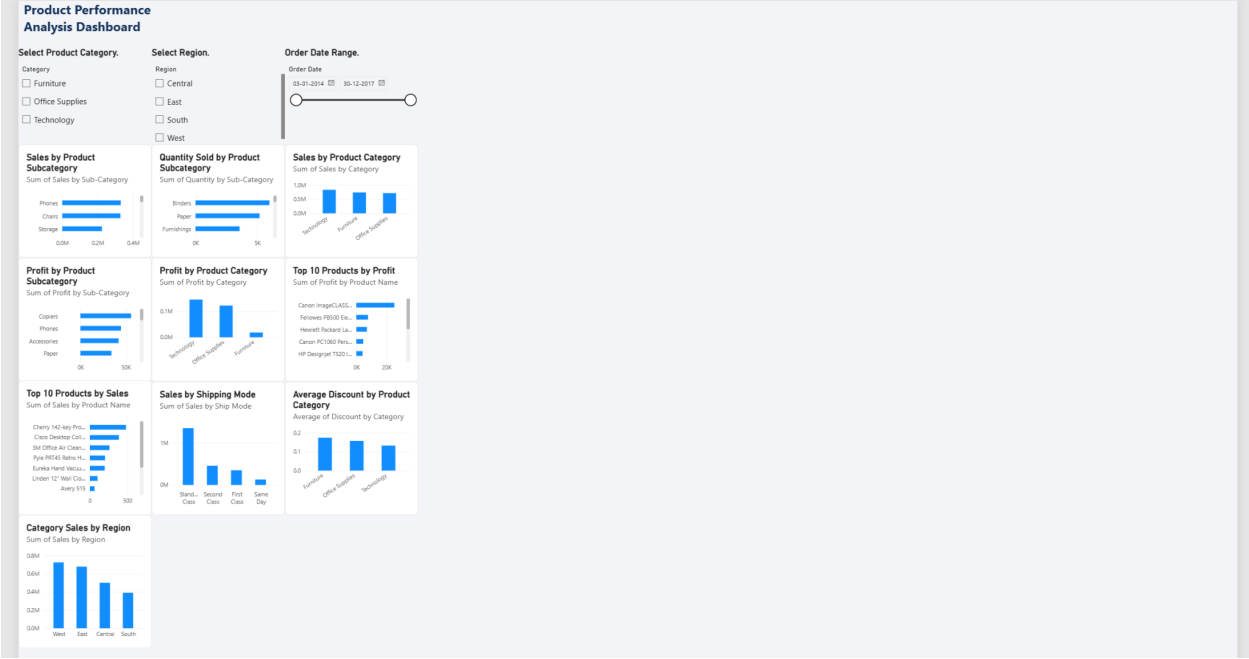
Page 2 — Customer Analysis



This page shows average sales per customer, unique customers, date/region/segment slicers, top customers, sales and orders by segment, top cities, segment profit and segment sales contribution. The screenshot shows 793 customers and average sales per customer displayed as 2.90K; the workbook-derived value is $2,896.85.

Layout improvement: expand visuals across the page and improve the readability of the donut labels and legend. Format average sales per customer as currency with two decimals.

Page 3 — Product Performance Analysis



This page combines sub-category sales, profit and quantity; category sales and profit; top products by sales and profit; shipping-mode sales; average discount by category; and category sales by region.

Layout improvement: the screenshot shows a large unused area on the right. Reflow into a balanced grid or reduce canvas width, and position the final regional chart deliberately.

Key business insights

TOPIC	EVIDENCE	BUSINESS MEANING
Category	Technology sales $836,154.03; profit $145,454.95.	Strongest combined revenue and profit category.
Region	West sales $725,457.82; profit $108,418.45.	West leads both measures.
Segment	Consumer sales $1,161,401.34; profit $134,119.21.	Largest segment; protect and grow its base.
Shipping	Standard Class sales $1,358,215.74.	Dominant mode; assess cost and service levels.
Sub-category	Tables $-17,725.48; Bookcases $-3,472.56; Supplies $-1,189.10 profit.	Investigate negative product economics.
Top product	Canon imageCLASS 2200 Advanced Copier — $61,599.82 sales.	Monitor availability of a major revenue contributor.
Top profit product	Canon imageCLASS 2200 Advanced Copier — $25,199.93 profit.	Leading product-level profit contributor.
Peak month	2017-11 — $118,447.82 sales.	Use trend to inform seasonal planning.

Business recommendations

- Protect Technology category performance through availability planning, supplier reliability and targeted cross-selling.
- Use West as a benchmark; investigate Central and South performance by product mix, customer mix and local demand.
- Prioritise Consumer retention while designing relevant propositions for Corporate and Home Office customers.
- Review Tables, Bookcases and Supplies for discount depth, pricing, sourcing, shipping cost and product returns.
- Test whether discounts generate incremental profitable sales rather than only higher volume.
- Use monthly patterns to plan inventory and campaigns, while validating seasonality over multiple years.
- Compare Standard Class fulfilment cost, delivery performance and satisfaction with faster shipping options.

Conclusion and future scope

The project demonstrates a complete analytics workflow: source-data inspection, Excel exploratory analysis, SQL query development, KPI definition and interactive Power BI reporting. It converts transactional records into actionable views of category, region, customer, product and shipping performance.

- Sales forecasting with seasonality validation.
- Customer retention and repeat-purchase cohort analysis.
- RFM-based customer segmentation.
- Discount and product contribution optimisation.
- Scheduled refresh and automated reporting.

Technical validation notes

- Excel source: 9,994 rows. SQL script: 29 SELECT statements. No SQL execution output was included.
- Reconcile MySQL COUNT(*) with the 9,994-row Excel source before claiming cross-tool consistency. Check import settings, skipped rows and date parsing.
- The PBIX file was supplied, but the internal model and DAX expressions were not extracted for this report.
- Screenshot KPI values align with workbook totals after rounding; the profit card displays abbreviated K notation.
- Confirm the Sales by Shipping Mode visual uses SUM of Sales, then refine the unused canvas space on all three pages.

Resume-ready project description

E-Commerce Sales and Customer Analytics | Excel, MySQL, Power BI

Analysed 9,994 retail transaction records to evaluate sales, profit, customer segments, product performance and regional trends. Built Excel pivot-table analysis, authored SQL queries using aggregations, grouping, distinct counts, date functions and ranking, and developed a three-page Power BI dashboard with KPI measures and interactive slicers. Identified Technology as the leading category, West as the top region, Consumer as the largest segment, and loss-making sub-categories including Tables and Bookcases.